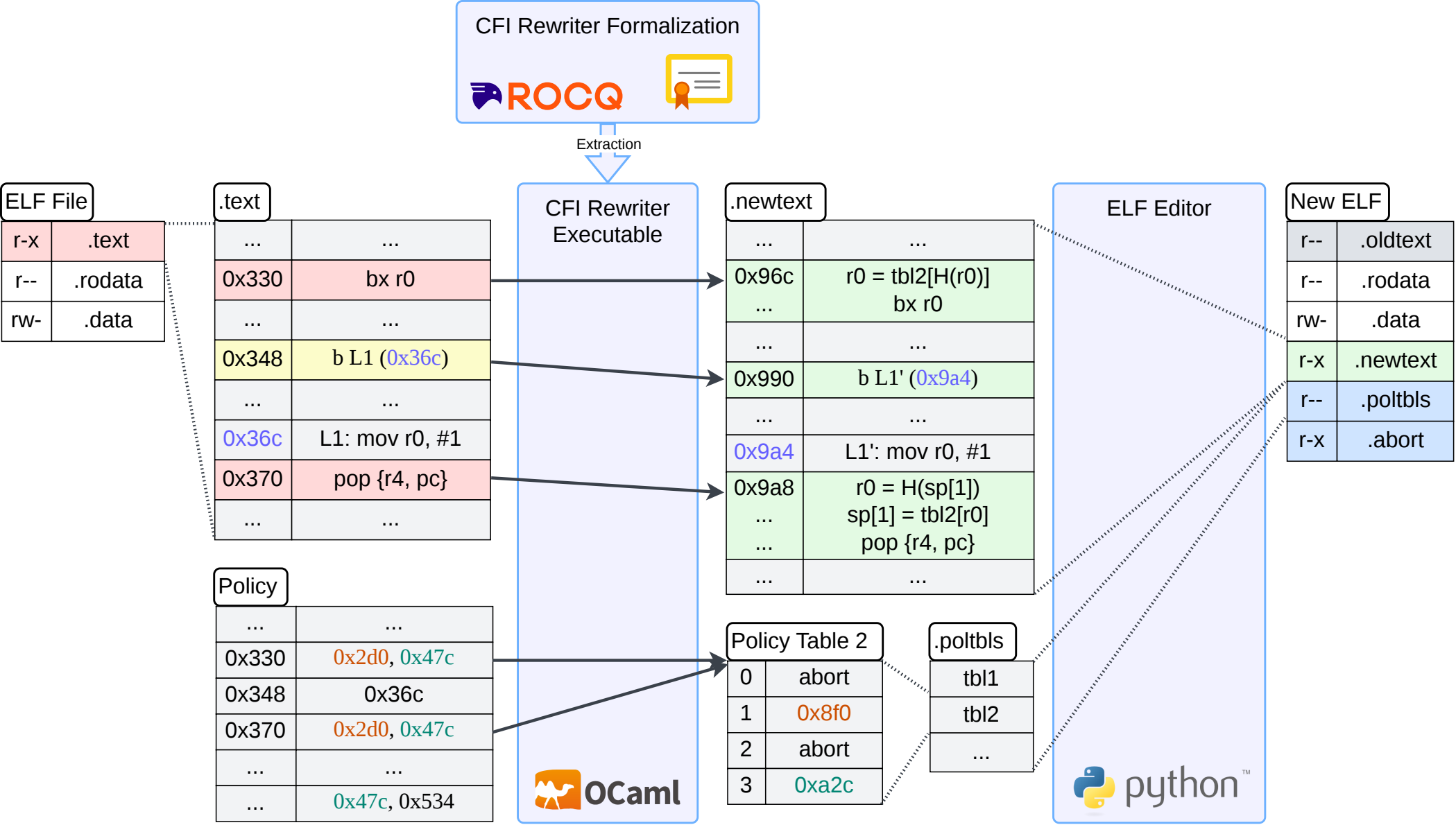

CFI Rewriter Formalization
ROCQ
Extraction
ELF File
r-x .text
r-- .rodata
rw- .data
.text
... ...
0x330 bx r0
... ...
0x348 b L1 (0x36c)
... ...
0x36c L1: mov r0, #1
0x370 pop {r4, pc}
... ...
CFI Rewriter Executable
OCaml
.newtext
... ...
0x96c r0 = tbl2[H(r0)]
... bx r0
... ...
0x990 b L1' (0x9a4)
... ...
0x9a4 L1': mov r0, #1
0x9a8 r0 = H(sp[1])
... sp[1] = tbl2[r0]
... pop {r4, pc}
... ...
Policy
... ...
0x330 0x2d0, 0x47c
0x348 0x36c
0x370 0x2d0, 0x47c
... ...
... 0x47c, 0x534
Policy Table 2
0 abort
1 0x8f0
2 abort
3 0xa2c
.poltbls
tbl1
tbl2
...
ELF Editor
python
New ELF
r-- .oldtext
r-- .rodata
rw- .data
r-x .newtext
r-- .poltbls
r-x .abort